
Checking Quran Recitation from Noisy ASR Transcripts: A Competition with Agent-Assisted Annotation

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Abstract

1 We propose two tasks for checking Quran recitation from ASR: detecting and
2 splitting ayahs, and labeling localized transcript/reference events. Task A has a
3 public 258-recording benchmark. For Task B, we completed a private annotation set
4 of 100 human-approved recording cases comprising 314 ayah chunks. Algorithms
5 receive reviewed splits and references. An agent track will annotate a separate
6 unlabelled development pool and build an algorithm for private evaluation. We
7 report annotation findings, distinguish the legacy machine-labeled pilot from the
8 completed gold review, and document input exposure that prevents claiming an
9 unseen-input test.

10 1 Task and motivation

11 Quran memorization involves a listener checking recitation. An ASR-based application must handle
12 opening formulas, repeats, restarts, repairs, and missing or substituted words. Edit distance alone
13 does not distinguish these cases. Our labels assess transcript/reference differences, not whether the
14 speaker or ASR caused them; unwritten vowel and tajweed errors are outside scope.

15 In **Task A (detect and split)**, a full transcript maps to ayah IDs and word assignments, or abstention
16 on non-Quran input. The public v1.1 benchmark exists. In **Task B (event annotation)**, outputs
17 have combined labels such as `substitution_mistake` and `repetition_benign`, with hypothe-
18 sis/reference spans. Review covers whole recordings and opening formulas. Algorithms receive
19 reviewed chunks, ayah IDs and exact references; scoring excludes detection errors.

20 The agent track supplies unlabelled development transcripts, the Quran reference, definitions and
21 disjoint examples. Entrants create annotations and an algorithm; final code is scored. Agreement
22 with self-generated labels is not independent annotation accuracy. Gold performance assesses the
23 combined process, not every development annotation or annotation’s causal contribution.

24 2 Data and findings from human review

25 The public Task A release has 258 transcripts: 254 Quranic recordings, four non-Quran negatives and
26 1,267 reviewed chunks covering 604 ayahs from 45 surahs. Its train/test split has 151/107 recordings
27 and 711/556 chunks. This is a public benchmark; duplicates and related inputs must stay in one
28 partition.

29 **Completed Task B review.** As of 8 September 2026, all **100 selected recording cases** are human-
30 approved: 314 chunks, 233 no-event chunks, 115 within-ayah events, and 47 opening-formula
31 annotations. Three approved reserves are outside the scored set. The selection covers 274 ayahs in

32 38 surahs and ten learner IDs; one learner contributes 46 recordings. These purposive counts do not
33 estimate prevalence or imply unseen-reciter generalization. The rubric is v0.19; the evaluator and
34 metric are not frozen.

35 We verified recording IDs and transcripts against both source exports. Punctuation-normalized dupli-
36 cates and redundant clean subpassages are excluded. Shared passages remain only for documented
37 differences in event pattern or clean/error contrasts. Distinct IDs and paths do not establish different
38 audio takes.

39 The review settled four rules. **Omissions** include missing beginnings and endings, with no be-
40 nign exemption for a final ayah. Whole-ayah omissions require a reviewed continuous-passage
41 interpretation; a completed final ayah does not imply future omissions. **Repeats** span the original
42 phrase and its extra copy against one reference copy. Wrong-then-correct attempts remain corrected
43 events; a correct-then-incorrect restatement is `substitution_mistake` spanning both attempts.
44 **Spelling** differences handled by agreed normalization receive no event; contextual residual forms
45 use `spelling_benign`, with wasl explained in the note. This permits no blanket final-letter deletion.
46 **Localization** uses verbatim words and half-open spans in original token arrays, including empty
47 hypothesis spans for omissions. Merge continuous replacements, but preserve separate events across
48 matching text or attempt boundaries.

49 Opening formulas and complete letter-name representations receive benign labels. Possible ASR
50 origin does not create an uncertain verdict.

51 3 Evaluation and feasibility evidence

52 The existing Task A scalar sums detection, split, and abstention errors. Earlier work reported held-out
53 scores of 0.760 for the incumbent and 0.079 for the best of six coding-agent runs.¹

54 The legacy Task B pilot has 99 machine-labeled chunks from 20 recordings, pending human review
55 under its old schema. Its public-test scores were 2.000 for an empty baseline and 1.111 for naive diff
56 and the incumbent. Its separate type/verdict fields, uncertainty category and superseded rules do not
57 validate the new gold. The documented export mismatch—1,268 Task B records versus 1,267 Task A
58 chunks—is preserved in the historical release.

59 For the new protocol, freeze matching, scalar weights, the supplied-split interface, budgets, and
60 hypotheses before runs. Report label and localization quality, missed mistakes, false flags on
61 clean/benign/corrected cases, and per-label results. Matching must handle omission anchors and
62 both-attempt spans, and prevent one broad prediction from earning credit for several distinct gold
63 events. Re-run baselines under the frozen contract; no new Task B algorithm results are claimed.

64 4 Competition plan and exposure controls

65 Protect the completed gold snapshot and prepare a separate unlabelled development bundle, excluding
66 selected inputs and duplicate-equivalent material. Supply disjoint or synthetic examples and a fixed
67 clarification exercise before measured runs. Agents receive no gold answers or gold-score feedback;
68 final code is evaluated privately. Exclude reviewer files, source repositories, answer-bearing history,
69 and annotation-session memory from agent runtimes.

70 **Exposure limitation.** All 100 selected transcripts match public Task A inputs; twelve also match
71 transcripts represented in the public legacy Task B labelled pilot. New adjudications and selected
72 membership remain private, but the inputs are not unpublished. Runtime isolation must exclude
73 public-release retrieval and cannot itself establish absence of prior exposure. An unseen-input final
74 competition ranking requires a separate collection of previously unreleased recordings, with reuse
75 permission and de-identification verified before launch. That collection is planned, not already
76 available.

77 Shared docs contain aggregate findings and synthetic examples; gold answers stay in a separate
78 private reviewer bundle. Verify privacy and reuse scope before new publication.

¹Autoresearch with coding agents: generalizers and metric-maximizers on Quran recitation data, AIST 2026. These are earlier Task A results, not Task B experiments.